NJALA UNIVERSITY

DEPARTMENT OF BSC NURSING

BO CAMPUS

HISTOPATHOPHYSIOLOGY NOTE

Blood Dyscrias

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Blood Dycrias

A. Bone Marrow and Hematopoiesis

Although the mature formed elements of blood are quite different from each other in both structure and function, all of these cells develop from a common hematopoietic **stem cell** population, which resides in the bone marrow. The developmental process is called **hematopoiesis** and represents an enormous metabolic task for the body. More than 100 billion cells are produced every day. This makes the bone marrow one of the most active organs in the body. In adults, most of the active marrow resides in the vertebrae, sternum, and ribs. In children, the marrow is more active in the long bones.

Cytokines that regulate hematopoiesis

Cytokine	Cell Lines Stimulated	Cytokine Source
IL-1	Erythrocyte Granulocyte Megakaryocyte Monocyte	Multiple cell types
IL-3	Erythrocyte Granulocyte Megakaryocyte Monocyte	T lymphocytes
IL-4	Basophil	T lymphocytes
IL-5	Eosinophil	T lymphocytes
IL-6	Erythrocyte Granulocyte Megakaryocyte Monocyte	Endothelial cells Fibroblasts Macrophages
IL-11	Erythrocyte Granulocyte Megakaryocyte	Fibroblasts Osteoblasts
Erythropoietin	Erythrocyte	Kidney Kupffer cells of liver
SCF	Erythrocyte Granulocyte Megakaryocyte Monocyte	Multiple cell types
G-CSF	Granulocyte	Endothelial cells Fibroblasts Monocytes
GM-CSF	Erythrocyte Granulocyte Megakaryocyte	Endothelial cells Fibroblasts Monocytes T lymphocytes
M-CSF	Monocyte	Endothelial cells Fibroblasts Monocytes
Thrombopoietin	Megakaryocyte	Liver, kidney

Key: IL, interleukin; CSF, colony-stimulating factor; G, granulocyte; M, macrophage; SCF, stem cell factor.

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COAGULATION FACTORS & THE COAGULATION SYSTEM

The coagulation system provides for immediate activation when control of bleeding (**hemostasis**) is required and confines its activity to the site of blood loss. Otherwise, coagulation might occur throughout the entire circulatory system, which would be incompatible with life.

The major components of hemostasis are platelets, endothelial cells (lining the blood vessels), other tissue factor (TF)–bearing cells and the coagulation factors, which are plasma proteins. The end result of the activated coagulation system is the formation of a complex of cross-linked **fibrin** molecules and platelets that terminate hemorrhage after injury.

All of the factors have roman numerals, and the inactive forms are written without annotation (eg, factor II, also known as prothrombin). The activated forms of the factors are signified by the letter “a” (eg, factor IIa, also known as thrombin).

Most of the coagulation factors are made by the liver, but factor XIII derives from platelets and factor VIII is made by endothelial cells. Factors II, VII, IX, and X are particularly important factors (Table 6-3) because they are all dependent on the liver enzyme γ-carboxylase. Gammacarboxylase is dependent on vitamin K, and the oral anticoagulant **warfarin** acts by interfering with vitamin K activity. Two of the anticoagulant proteins, protein S and protein C (see later discussion), are also vitamin K dependent.

Coagulation factors of plasma.

Name	Production Source
Procoagulant factors	
Factor I (fibrinogen)	Liver
Factor II (prothrombin)	Liver
Factor III (tissue thromboplastin)	Tissue
Factor IV (calcium)	...
Factor V (proaccelerin)	Liver
Factor VI (obsolete = factor Va)	...
Factor VII (proconvertin)	Liver
Factor VIII (antihemophilic factor)	Endothelial cells
Factor IX (Christmas factor)	Liver
Factor X (Stuart-Prower factor)	Liver
Factor XI (plasma thromboplastin antecedent)	Liver
Factor XII (Hageman factor)	Liver
Factor XIII (fibrin-stabilizing factor)	Platelets
Anticoagulant factors	
Antithrombin	Liver
Protein C	Liver
Protein S	Liver
Plasminogen	Liver
Tissue factor pathway inhibitor	Endothelial cells

OVERVIEW OF BLOOD DISORDERS

FORMED ELEMENT DISORDERS

Disorders of red cells, white cells, and platelets are separated for discussion because one or the other is found to be the most abnormal during laboratory testing. However, because of the clonal nature of hematopoiesis, many disorders affect all the formed elements of the blood.

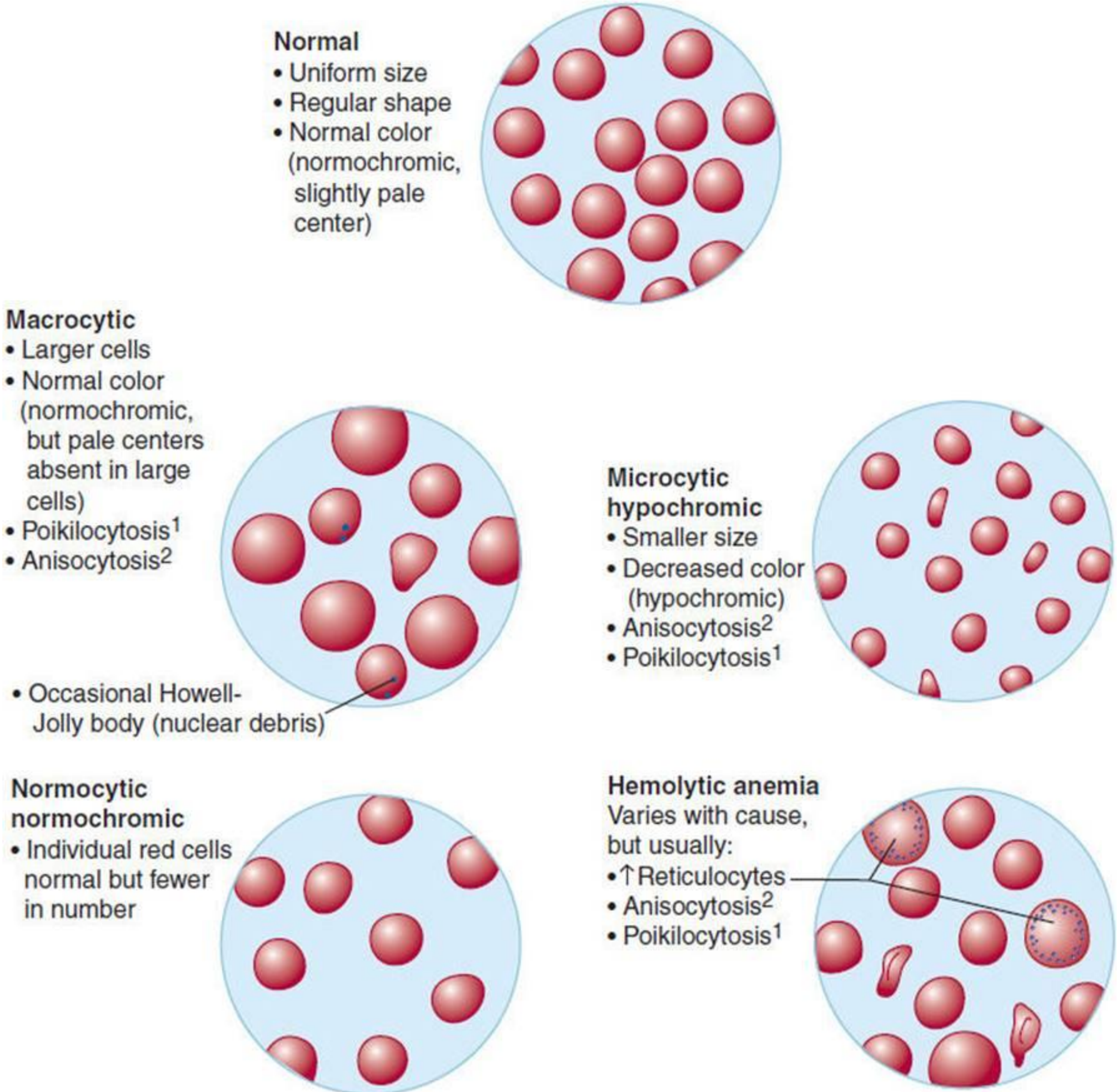
1. Red Cell Disorders

There are many red cell abnormalities, but the principal ones are a variety of anemias. **Anemia** is defined as an abnormally low hemoglobin concentration in the blood. There are several methods of classification, but the prevailing systems are based on red cell size and shape.

In normal persons, erythrocytes are of uniform size and shape, and the automated blood count shows a mean corpuscular volume (MCV) near 90 fL, which is the estimated volume of a single cell. Automated systems usually report abnormalities of red cells as changes in hemoglobin concentration, red cell number, and MCV. Small cells (with low MCVs) are termed **microcytic**, and cells larger than normal are termed **macrocytic**. The relative nonuniformity of cell shapes (**poikilocytosis**) or sizes (**anisocytosis**) can further aid in subclassifying erythrocyte disorders.

TABLE 6-4 Morphologic classification and common causes of anemia.

Type	MCV	Common Causes
Macrocytic	Increased	Folic acid deficiency
		Vitamin B_{12} deficiency
		Liver disease
		Alcohol
		Hypothyroidism
		Drugs (sulfonamides, zidovudine, antineoplastic agents)
		Myelodysplastic syndromes
Microcytic	Decreased	Iron deficiency
		Thalassemias
Normocytic	Normal	Aplastic anemia
		Anemia of chronic disease
		Chronic kidney disease
		Hemolytic anemia
		Spherocytosis



The macrocytic anemias reflect either abnormal nuclear maturation or a higher fraction of young, large red cells (reticulocytes). When the nuclei of maturing red cells appear too young and large for the amount of hemoglobin in the cytoplasm, the macrocytic anemia is termed **megaloblastic**. These anemias are most often due either to vitamin deficiencies (vitamin B_{12} or folic acid) or drugs that interfere with DNA synthesis.

The normocytic anemias can be due to multiple causes: decreased numbers of red cell precursors in the marrow (primary failure called aplastic anemia, replacement of marrow elements with cancer, certain viral infections, or autoimmune inhibition called **pure red cell aplasia**), low levels of erythropoietin (resulting from chronic kidney disease), or chronic inflammatory diseases that affect the availability of iron in the marrow. Other normocytic anemias can be secondary to decreased life span of the cells that are produced. Examples of this phenomenon are acute blood loss; **autoimmune hemolytic anemias**, in which antibodies or complement bind to red cells and cause their destruction; **sickle cell anemia**, in which the abnormal hemoglobin polymerizes and obliterates the usual resilience of the red cell; and **hereditary spherocytosis** or **hereditary elliptocytosis**, in which defects in the erythrocyte membrane affect their ability to squeeze through the capillary microcirculation.

Anemias are very common. In contrast, an elevated hemoglobin concentration, termed **erythrocytosis**, is uncommon. Elevations in hemoglobin concentration can occur as a secondary phenomenon because of increased erythropoietin levels, such as that found in smokers or people who live at high altitudes (whose low blood oxygen levels stimulate erythropoietin production). Some tumors, especially renal tumors, can also make erythropoietin. Primary **polycythemia** is an abnormality of the bone marrow itself. This myeloproliferative syndrome leads to an increased red cell mass and consequent low erythropoietin levels by the negative-feedback mechanism discussed previously.

2. White Blood Cell Disorders

Abnormalities in white cell numbers occur commonly , whereas abnormalities of function are rare. Neoplastic transformation in the form of leukemia (granulocytes and monocytes) or lymphoma (lymphocytes) is fairly common.

TABLE 6-5 Causes of abnormal neutrophil counts.

Changes in neutrophil count are the most common white cell abnormality detected on the automated blood count. Increased numbers of neutrophils (**leukocytosis**) suggest acute or chronic infection or inflammation but can be a sign of many conditions. These include stress, because adrenal corticosteroids cause **demargination** of neutrophils from blood vessel walls.

Decreased numbers of neutrophils (**neutropenia**) can be seen in overwhelming infection and benign diseases such as **cyclic neutropenia** (see later discussion) but can also be seen when the bone marrow is infiltrated with tumor or involved by the myelodysplastic syndromes. Many drugs can also directly suppress marrow production, and because neutrophils have the shortest half-life in the blood of any cell produced by the marrow, their numbers may fall quickly.

Lymphocyte counts are classically elevated in viral infections, such as infectious mononucleosis. However, persistent elevations suggest malignancies, particularly **chronic lymphocytic leukemia**, which may not cause any symptoms and be incidentally discovered on a routine blood count.

Neutrophilia	Neutropenia
Increased marrow activity	*Decreased marrow activity*
Bacterial infections	Drugs (antineoplastic agents, antibiotics, gold, certain diuretics, antithyroid agents, antihistamines, antipsychotics)
Acute inflammation	Radiation exposure
Leukemia and myeloproliferative disorders	Megaloblastic anemia
Release from marrow pool	Cyclic neutropenia
Stress (catecholamines)	Kostmann (infantile) neutropenia
Corticosteroids	Aplastic anemias
Endotoxin exposure	Myelodysplastic syndromes
Demargination into blood	Marrow replacement by tumor
Bacterial infections	*Decreased neutrophil survival*
Hypoxemia	Sepsis
Stress (catecholamines)	Viral or rickettsial infection
Corticosteroids	Immune destruction associated with drugs
Exercise	Immune destruction associated with autoantibodies (systemic lupus erythematosus, Felty syndrome)
	Hypersplenism

Causes of abnormal lymphocyte counts.

Lymphocytosis
Medium to large, atypical lymphocytes predominant
Viral infections (mononucleosis, mumps, measles, hepatitis, rubella)
Active immune responses, particularly in children
Toxoplasmosis
Lymphoma with circulating cells
Chronic lymphocytic leukemia
Small, mature lymphocytes predominant
Chronic infections (tuberculosis)
Autoimmune diseases (myasthenia gravis)
Metabolic diseases (Addison disease)
Lymphoma with circulating cells
Chronic lymphocytic leukemia
Immature cells predominant
Acute lymphocytic leukemia
Lymphoblastic lymphoma
Lymphopenia
Immunodeficiency states (AIDS)
Corticosteroid therapy
Toxic drugs
Cushing syndrome

Decreased lymphocyte counts (**lymphopenia**) are a common complication of corticosteroid therapy but are most worrisome for immunodeficiency states; HIV directly infects lymphocytes, and the likelihood of opportunistic infections increases as lymphocyte counts fall, resulting in AIDS.

3. Platelet Disorders

Abnormalities in platelet number are fairly common, particularly low counts (**thrombocytopenia**). Causes are listed in Table 6-7. Decreased production of platelets occurs when the marrow is affected by a variety of diseases or when TPO production by the liver is impaired, as in cirrhosis. Increased destruction of platelets is much more prevalent. There are three general mechanisms. Because a significant number of platelets normally reside in the spleen, any increase in spleen size or activity (**hypersplenism**) leads to lower platelet counts. Platelet consumption that is due to ongoing clotting will also lower counts. Most commonly, however, there is immune-mediated consumption caused by either drugs or autoantibodies. The latter are usually directed against the platelet membrane antigen gpIIb/IIIa.

TABLE 6-7 Causes of platelet abnormalities.

Thrombocytosis
Myeloproliferative disorders, especially essential thrombocythemia
Postsplenectomy
Reactive (postsurgical, posthemorrhage, anemias)
Inflammatory disorders
Malignancies
Thrombocytopenia
Decreased production
Aplastic anemia
Marrow infiltration
Vitamin B_{12} and folate deficiencies
Radiation or chemotherapy
Hereditary
Infection (HIV, parvovirus, CMV)
Cirrhosis (low thrombopoietin levels)
Decreased survival
Immune mediated (idiopathic, systemic lupus erythematosus, drug induced, neonatal from maternal IgG)
Hypersplenism
Disseminated intravascular coagulation
Thrombotic thrombocytopenic purpura, hemolytic uremic syndrome
Prosthetic valves
Qualitative platelet disorders
Inherited
Bernard-Soulier syndrome (adhesion defect)
Glanzmann thrombasthenia (aggregation defect)
Storage pool disease (granule defect)
Von Willebrand disease
Wiskott-Aldrich syndrome
Acquired
Uremia
Dysproteinemias
Chronic liver disease
Drug induced (especially aspirin)

Functional platelet disorders are common, especially the acquired disorders resulting from uremia (renal failure) or aspirin, which inhibits the platelet enzyme cyclooxygenase and decreases platelet aggregability. Inherited abnormalities are unusual with the exception of **von Willebrand disease**, which results from either quantitative or qualitative defect of vWF, the carrier protein for factor VIII. vWF also acts as a bridge between platelets and the endothelium and thus is crucial for formation of the platelet plug in the coagulation cascade.

Elevations in the platelet count above normal (**thrombocytosis**) are relatively common and are especially apt to occur in recovery from iron deficiency anemia upon iron repletion. In the myeloproliferative disorders, such as polycythemia, platelet counts are often high. In **essential thrombocythemia**, platelet counts may be higher than 1,000,000/µL.

COAGULATION FACTOR DISORDERS

The most important coagulation factor disorders are quantitative rather than qualitative and usually hereditary rather than acquired . Exceptions to this rule are **acquired factor inhibitors**, which are antibodies that bind to one of the coagulation factors, most often factor VIII. These may or may not cause clinical bleeding problems, but they can be extremely difficult to treat. The quantitative disorders that most commonly cause bleeding are **hemophilia A** (deficiency of factor VIII) and **hemophilia B** (deficiency of factor IX). Both are X chromosome–linked recessive traits, and affected males have very low levels of factor VIII or IX. It is not clear why all affected males do not have complete absence of factor VIII or IX activity. Hemophilia A is more common, with a prevalence of 1:10,000 males worldwide. Both disorders lead to spontaneous and excessive post-traumatic bleeding, particularly into joints and muscles. Females with the trait have 50% of the normal amount of either factor and tend not to have any bleeding problems; in general, one needs only half of the normal quantities of most coagulation factors to clot normally. The aPTT test is usually designed to become abnormal when factor VIII or IX activities fall below 50% of normal.

TABLE 6-8 Coagulation factor deficiencies.

Factor	Disease	Inheritance Pattern	Frequency	Disease Severity
Fibrinogen	Afibrinogenemia	Autosomal recessive	Rare	Variable
	Dysfibrinogenemia	Autosomal dominant	Rare	Variable
Factor V	Parahemophilia	Autosomal recessive	Very rare	Moderate to severe
Factor VII		Autosomal recessive	Very rare	Moderate to severe
Factor VIII	Hemophilia A	X-linked recessive	Common	Mild to severe
vWF	von Willebrand disease	Autosomal dominant	Common	Mild to moderate
Factor IX	Hemophilia B	X-linked recessive	Uncommon	Mild to severe
Factor X		Autosomal recessive	Rare	Variable
Factor XI	Rosenthal syndrome	Autosomal recessive	Uncommon	Mild
Factor XII	Hageman trait	Autosomal recessive or dominant	Rare	Asymptomatic
Factor XIII		Autosomal recessive	Rare	Severe

Vitamin K deficiency also leads to quantitative declines in the levels of factors II, VII, IX, and X and proteins C and S; prolongation of the prothrombin time may result.

Quantitative inherited abnormalities of the anticoagulation systems also occur. Protein S deficiency, protein C deficiency, and AT deficiency all occur and lead to abnormal clotting problems, as discussed in the next section.

Finally, the condition of **consumptive coagulopathy** or **disseminated intravascular coagulation** (**DIC**) needs to be included. This condition is generally due to overwhelming infection, specific leukemias or lymphomas, or massive hemorrhage. In DIC, the coagulation factors become depleted. Often there is simultaneous activation of the fibrinolytic system as well, and uncontrolled bleeding may occur throughout the entire circulatory system. PT and aPTT are usually both abnormal.